

If $[H, \Pi] = 0$, and $|n\rangle$ non-degenerate H eigenstate, then $|n\rangle$ is parity eigenstate.

ex) SHO

$$a^\dagger \sim \hat{x} - i\hat{p} \text{ so parity odd, } \Pi a^\dagger \Pi = -a^\dagger$$

Then

$$\Pi a^\dagger |0\rangle = -|1\rangle$$

In general,

$$\Pi |n\rangle = (-1)^n |n\rangle$$

Hydrogen

$$\text{g.s. } |1,0,0\rangle = |1,0,0\rangle$$

$$\langle \vec{r} | 1,0,0 \rangle = R_{1,0}(r) Y_0^0(\theta, \phi) = \left(\frac{2}{\pi a_0} \right)^{3/4} e^{-r/a_0} \frac{1}{\sqrt{4\pi}}, \text{ parity even}$$

1st excited: $|n, l, m\rangle = |2,0,0\rangle$ (degenerate in l, m)

Superposition: $|\psi\rangle = c_p |n=2, l=1\rangle + c_s |n=2, l=0\rangle$, not parity eigenstate.

Selection rules:

Two states $|\alpha\rangle, |\beta\rangle$ with $\Pi|\alpha\rangle = \epsilon_\alpha |\alpha\rangle, \Pi|\beta\rangle = \epsilon_\beta |\beta\rangle$

Dipole transition, α to β :

$$\begin{aligned} \langle \beta | x | \alpha \rangle &= \langle \beta | \Pi (\Pi^\dagger x \Pi) \Pi^\dagger | \alpha \rangle \\ &= \langle \beta | \epsilon_\beta (-x) \epsilon_\alpha | \alpha \rangle \\ &= -\epsilon_\alpha \epsilon_\beta \langle \beta | x | \alpha \rangle \end{aligned}$$

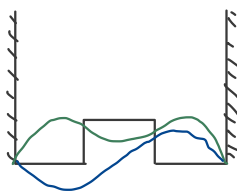
$$\Rightarrow \epsilon_\alpha \epsilon_\beta = -1, \text{ must have opposite parity}$$

Mann's rule:

Transition $\langle \psi_f | \mathcal{O} | \psi_i \rangle$, if f has parity ϵ_f (ϵ_f), $\mathcal{O}$ has parity $\epsilon_{\mathcal{O}}$

Requires: $\epsilon_f = \epsilon_{\mathcal{O}} \epsilon_i \Leftrightarrow \epsilon_f \epsilon_{\mathcal{O}} \epsilon_i = +1$.

ex) Double well



Left + right states:

$$|L\rangle = \frac{1}{\sqrt{2}} (|S\rangle + |A\rangle)$$

$$|R\rangle = \frac{1}{\sqrt{2}} (|S\rangle - |A\rangle)$$

Non-stationary, oscillate between L, R with freq. $\omega = \frac{E_0 - E_1}{\hbar}$